

Hepatocellular Carcinoma and Cirrhosis: A Review of their Relative Incidence in a 25-year Period in the Florence Area

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Summary

An eight-fold increase in the incidence of hepatocellular carcinoma in the Florence area was detected in a 25-year retrospective review of adult autopsy records in the Institute of Pathology of the University of Florence.

During the same period, the incidence of cirrhosis did not show a parallel increase.

The relationship between hepatocellular carcinoma, cirrhosis and HB virus is briefly discussed in the light of these findings.

Key-Words: Carcinoma — Cirrhosis — Incidences

The incidence of hepatocellular carcinoma (HCC) in a series of 27,286 consecutive adult autopsies performed at the Institute of Pathology at the University of Florence between 1887 and 1956, was 0.61% (U. Bettini and S. Dini, 1958). Since the incidence of HCC seems to be on the increase in Italy (A. Ottolenghi, 1966; L. Pagiario, 1982), in other European countries (R.N.M. Mac Sween, 1974), and in western countries in general (R.L. Peters, 1976), we decided to undertake a retrospective review of the necropsy records of the Institute of Pathology at the University Hospital of Florence over the 25-year period 1958-1982, in order to determine whether a similar increase was detectable in our population.

Results

Analysis of the necropsy records for the period 1958-1982 disclosed 113 cases of histologically proven hepatocellular carcinoma, out of a total of 11,799 autopsies, representing a total necropsy HCC incidence of 0.96%. The percentage rises to 1.33 computing the HCC incidence in the 8,514 adult autopsies (older than 18 years). Hepatocellular carcinoma was almost four times more common in males (78.7%) than in females (21.3%).

During the period considered, the HCC incidence in adult autopsies steadily rose to reach an eightfold increase over the initial level (1958-62 incidence 0.30%; 1978-82 incidence 2.56%); this trend was similar in both sexes (female 0.00% and 0.47%; male 0.30% and 0.2.09%) (Fig.1).

During the same period, the adult necropsy incidence of cirrhosis was 5.95% (male cases 380; female cases 127). These data are summarized in Table 1.

The carcinoma/cirrhosis relationship (Fig. 1) was examined as suggested by R.A. Burnett et al. (1978), i.e., by computing the TC/T index (cases of tumor associated with cirrhosis (TC) divided by total tumor cases (T), and the TC/C index (percentage of cirrhosis cases in which tumor super-vened).

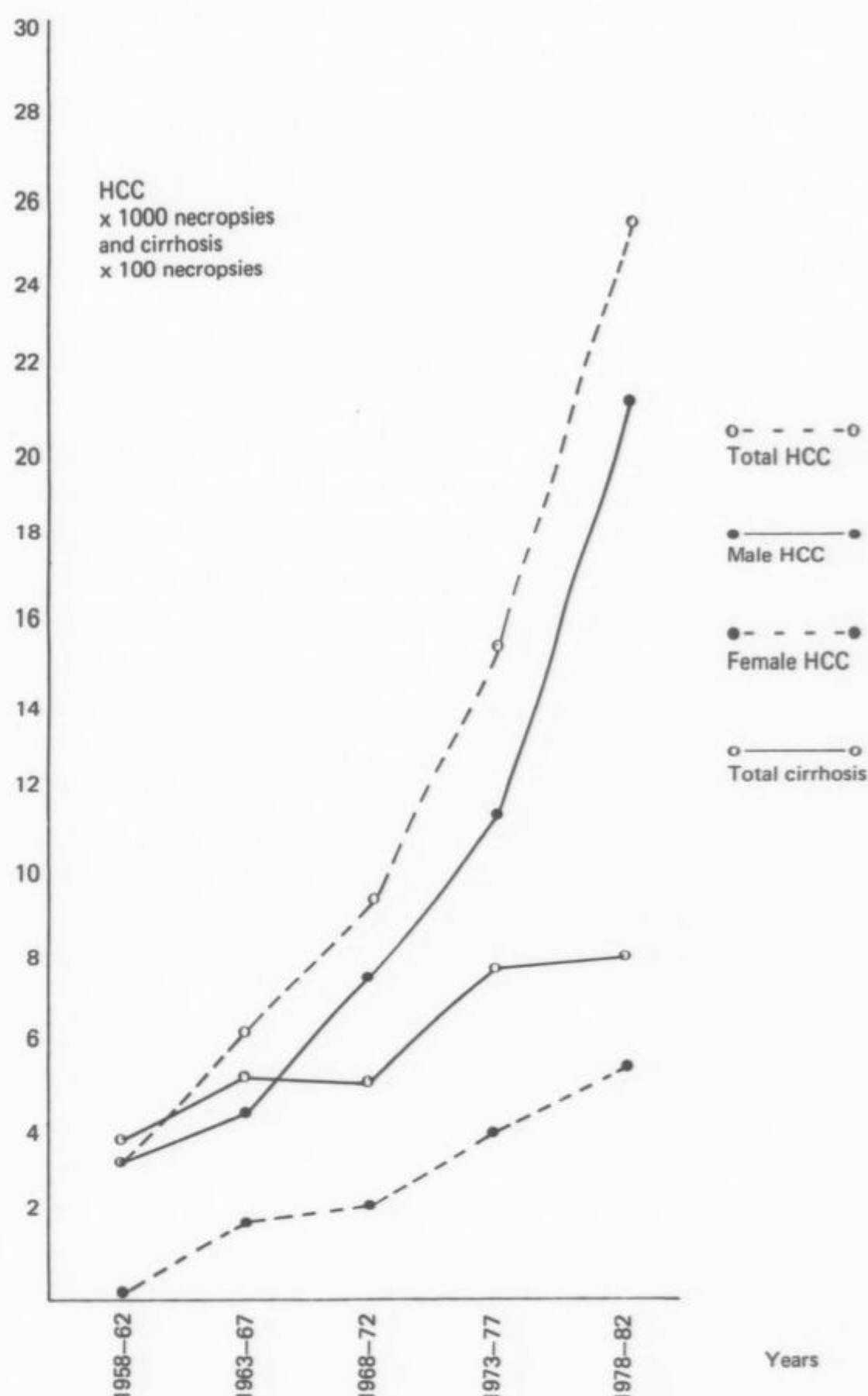


Fig. 1 Incidence of hepatic cirrhosis and hepatocellular carcinoma (HCC), over the 25-year period 1958-82, at the institute of pathology of the University of Florence

Table 1 Incidence of hepatic cirrhosis and HCC, with and without associated cirrhosis, in a total of 8.514 adult necropsies in the Institute of Pathology of Florence over the last 25 years (1958–1982)

	Total		M		F	
	No	%	No	%	No.	%
Cirrhosis	418	4.91	301	3.54	117	1.37
HCC	113	1.33	89	1.05	24	0.28
HCC with cirrhosis	89	1.05	79	0.93	10	0.12
HCC without cirrhosis	24	0.28	10	0.12	14	0.16

Table 2 HCC/cirrhosis relationship examined on the basis of TC/T, TC/C and male-female indices

Number of cases	TC/T		TC/T	
	No	%	No	%
Total	89/113	78.76	89/507	17.15
Male	79/89	88.88	79/380	20.78
Female	10/24	41.66	10/127	7.87

TC = HCC with cirrhosis; T = total HCC; C = cirrhosis + HCC with cirrhosis

Table 3 HCC/Cirrhosis relationship examined on the basis of TC/T and TC/C indices over the 25-year period 1958–1982

Years	TC/T		TC/T	
	No	%	No	%
1958–62	3/3	100	3/37	8.1
1963–67	5/8	62	5/67	7.4
1968–72	13/18	72	13/95	13.6
1973–77	30/35	85	30/168	17.8
1978–82	38/49	77	38/140	27.1

TC = HCC with cirrhosis; T = total HCC; C = cirrhosis+HCC with cirrhosis)

The overall TC/T index was 78.6% (88% for males and 41.66% for females) (Table 2). This index did not substantially change over the period under review (Table 3).

The overall TC/C index was 17.15% (20.78% for males and 7.87% for females) (Table 2). This index increased during the period under review from 8.1% in the first quinquennium to 27.1% in the last (Table 3).

Discussion

First of all it must be noted that while no change occurred in the racial composition of the general population in the area, the necropsy population might have been influenced by the fact that a specialised Division of Gastroenterology was established in Florence in 1972.

In spite of the limitations inherent in a retrospective necropsy survey, it seems clear that the data from our Institute are in agreement with other series from institutions from other countries that suggest a general increase in the incidence of hepatocellular carcinoma.

In particular, our series seems to be fairly comparable to that of *Burnett et al.*, which dealt with necropsy records of five Glasgow teaching hospitals over a 25-year period. In fact, our statistical data, both graphically and by percentage figures, are notably similar to those of the Scottish authors. Consequently, their findings and conclusions can be directly compared with ours.

During the period of review, the necropsy incidence of HCC

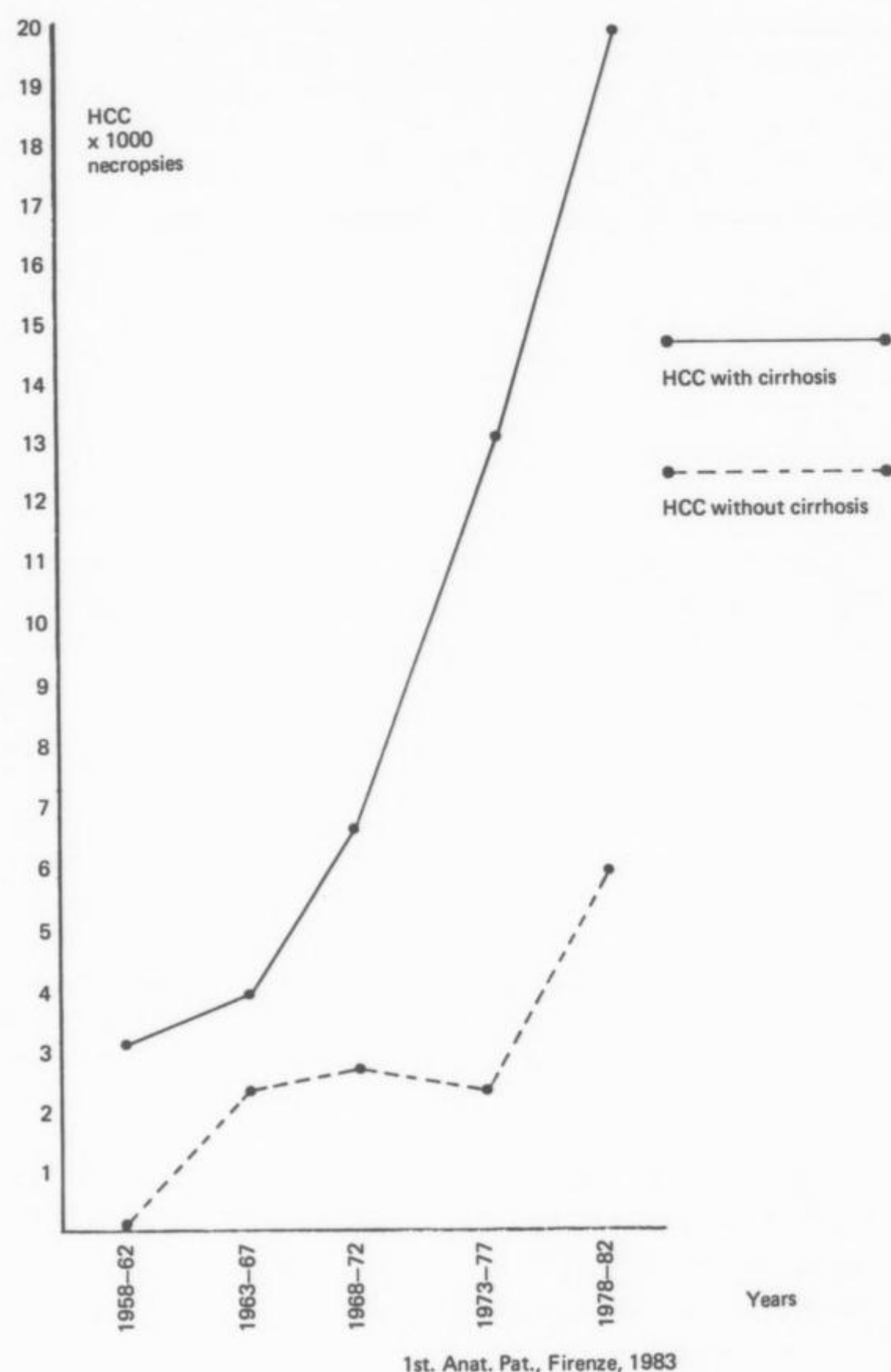
has steadily risen. This increase is not paralleled by an increase in cirrhosis; in fact, the cirrhosis curve shows only a modest rise in the last ten years examined.

Patton and Horn (1964) have suggested that the increase in the incidence of HCC might be due to longer survival in cirrhotic patients because of earlier diagnosis and improved treatment. Some support for this interpretation is the fact that the mean age at death of our HCC-cirrhosis associated patients (64 years) was slightly greater than in cirrhotics only (61 years). Also in the 25-year period reviewed we found a progressive increase in the percentage of the cirrhosis cases in which tumor supervened (8.1% incidence in 1958–62 to 27.1% incidence in 1978–82) (Table 3).

However, some data of our survey do not support this assertion.

We found that tumor increase was due to a parallel increase in HCC cases with and without associated cirrhosis (Fig. 2), we also found that, over the 25-year period, the mean age at death in both cirrhotic and HCC patients had not changed.

These observations, particularly the first, suggest the presence of an associated factor which might have expressed a more pronounced cirrhotogenic and/or carcinogenic effect, especially in the last ten years: this factor might be the HB virus. Such an hypothesis was already proposed by us in 1977 (*R. Iozzo, A. Giannini, F. Bartoloni St. Omer*), when we stated that "virus B could in and of itself lead to the



malignant transformation of the hepatocytes simultaneously giving place to a spectrum of alterations which can start from minimal necrotic or inflammatory lesions, sometimes even focal, up to chronic active hepatitis and cirrhosis”.

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◀ Fig. 2 Incidence of hepatocellular carcinoma (HCC), with and without cirrhosis associated, over the 25-year period 1958-82, in the Institute of Pathology of the University of Florence